

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS-ON EXPERIENCE (IOHE)**

**ON**

**AMOT: RANSOMWARE & FORENSICS ANALYSER**

**submitted in partial fulfilment of the requirements for the award of degree of**

**BACHELOR OF ENGINEERING**

**In**

**COMPUTER SCIENCE AND ENGINEERING**

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## **DECLARATION**

I hereby certify that the work which is being presented in the project report entitled “**AMOT:Ransomware & Forensics Analyser**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Preeti Saini**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place : Rajpura

**Aaryan Shyam**

Date : 14- May -2026

2210991121

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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## **Acknowledgement**

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I also thank my faculty members, peers, and friends for their valuable suggestions and assistance during the implementation and testing phases. Their feedback helped improve both the technical and analytical aspects of the project.

Finally, I would like to acknowledge the contribution of the open-source cybersecurity community, whose tools, frameworks, and research resources greatly supported the development of this ransomware detection and forensic analysis platform.

## **Abstract**

Ransomware attacks have become one of the most dangerous cybersecurity threats affecting organizations and individuals worldwide. Traditional antivirus systems rely heavily on signature-based detection methods, which are ineffective against modern polymorphic and zero-day malware. To overcome these limitations, this project presents AMOT: Ransomware & Forensics Analyser, an intelligent and automated malware detection platform.

The system integrates multiple detection mechanisms including static analysis, machine learning prediction, YARA rule matching, and OSINT-based threat intelligence verification into a unified analytical framework. Uploaded files are processed through a multi-stage pipeline that extracts file features, predicts malicious behaviour, scans for known malware signatures, and verifies threat reputation using public threat intelligence databases.

The backend of the system is developed using Python and Fast API, while the frontend interface is implemented using React.js. SQLite is used for storing historical analysis reports and metadata. The system generates detailed reports containing threat scores, malware classification, matched signatures, and mitigation recommendations.

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## **1. Introduction**

Cybersecurity threats are evolving rapidly, with ransomware attacks becoming increasingly sophisticated and difficult to detect. Modern malware uses encryption, obfuscation, and polymorphic behaviour to bypass traditional security systems. Conventional antivirus software depends mainly on signature-based detection, which is ineffective against unknown or newly emerging malware variants.

The AMOT: Ransomware & Forensics Analyser project aims to address these challenges by developing an intelligent and automated malware analysis platform capable of detecting both known and unknown threats. The platform combines machine learning algorithms, YARA signature matching, static analysis, and OSINT-based threat intelligence into a single unified framework.

The system helps cybersecurity analysts perform faster and more accurate malware investigations while reducing manual effort and alert fatigue. It generates comprehensive reports containing threat classifications, confidence scores, and mitigation recommendations.

### **1.1 Problem Statement**

Traditional antivirus systems face several limitations when dealing with modern ransomware threats. Signature-based detection techniques cannot effectively detect polymorphic malware or zero-day attacks. Organizations also face alert fatigue due to a large number of false-positive security alerts generated by existing security systems.

Manual malware analysis is time-consuming and requires expert knowledge. Security analysts often need to use multiple tools separately for static analysis, behavioural analysis, and threat intelligence verification. This creates delays in incident response and increases the risk of successful cyberattacks.

Therefore, there is a need for an intelligent, automated, and multi-layered malware analysis system capable of integrating multiple detection techniques into a single efficient platform.

## **1.2 Objectives of the Project**

The main objectives of the project are:

- To develop an intelligent ransomware detection and forensic analysis platform.
- To integrate static analysis, machine learning, YARA scanning, and OSINT verification into a single system.
- To reduce malware analysis time through automation.
- To improve detection accuracy and reduce false positives.
- To generate comprehensive and user-friendly forensic reports.
- To provide cybersecurity analysts with actionable insights and mitigation recommendations.
- To maintain historical analysis records for auditing and future investigation.

## **1.3 Proposed System**

The proposed system follows a multi-layered analytical pipeline where uploaded files are processed through multiple stages of detection and verification.

The system begins by securely receiving suspicious files uploaded by users. Static analysis is then performed to inspect the internal structure of the file without execution. Feature extraction techniques collect relevant file attributes which are passed to machine learning algorithms for malicious behaviour prediction.

The next stage uses YARA rule-based scanning to identify known ransomware signatures and suspicious patterns. Finally, the file's cryptographic hash is verified using public threat intelligence sources through OSINT APIs.

The outputs from all stages are combined to generate a final threat score and classification result.

## **2. Methodology**

The project methodology is divided into multiple analytical stages:

### **Step 1: File Upload and Preprocessing**

- User uploads suspicious file.
- File metadata and cryptographic hash are generated.
- File is stored securely for analysis.

### **Step 2: Static Analysis**

- File headers and metadata are inspected.
- Entropy calculation is performed.
- Suspicious strings and imports are extracted.
- Structural anomalies are detected.

### **Step 3: Machine Learning Analysis**

- Features are extracted using Pandas and NumPy.
- Trained machine learning model predicts malicious behaviour.
- Confidence score is generated.

### **Step 4: YARA Rule Matching**

- File is scanned using YARA rules.
- Known malware signatures and patterns are matched.
- Detection results are generated.

### **Step 5: OSINT Verification**

- File hash is sent to threat intelligence APIs.
- Reputation reports are collected.
- Threat verdict is generated.

### **Step 6: Results Fusion**

- Results from all modules are combined.
- Overall threat score is calculated.
- Final classification is determined.

### **Step 7: Report Generation**

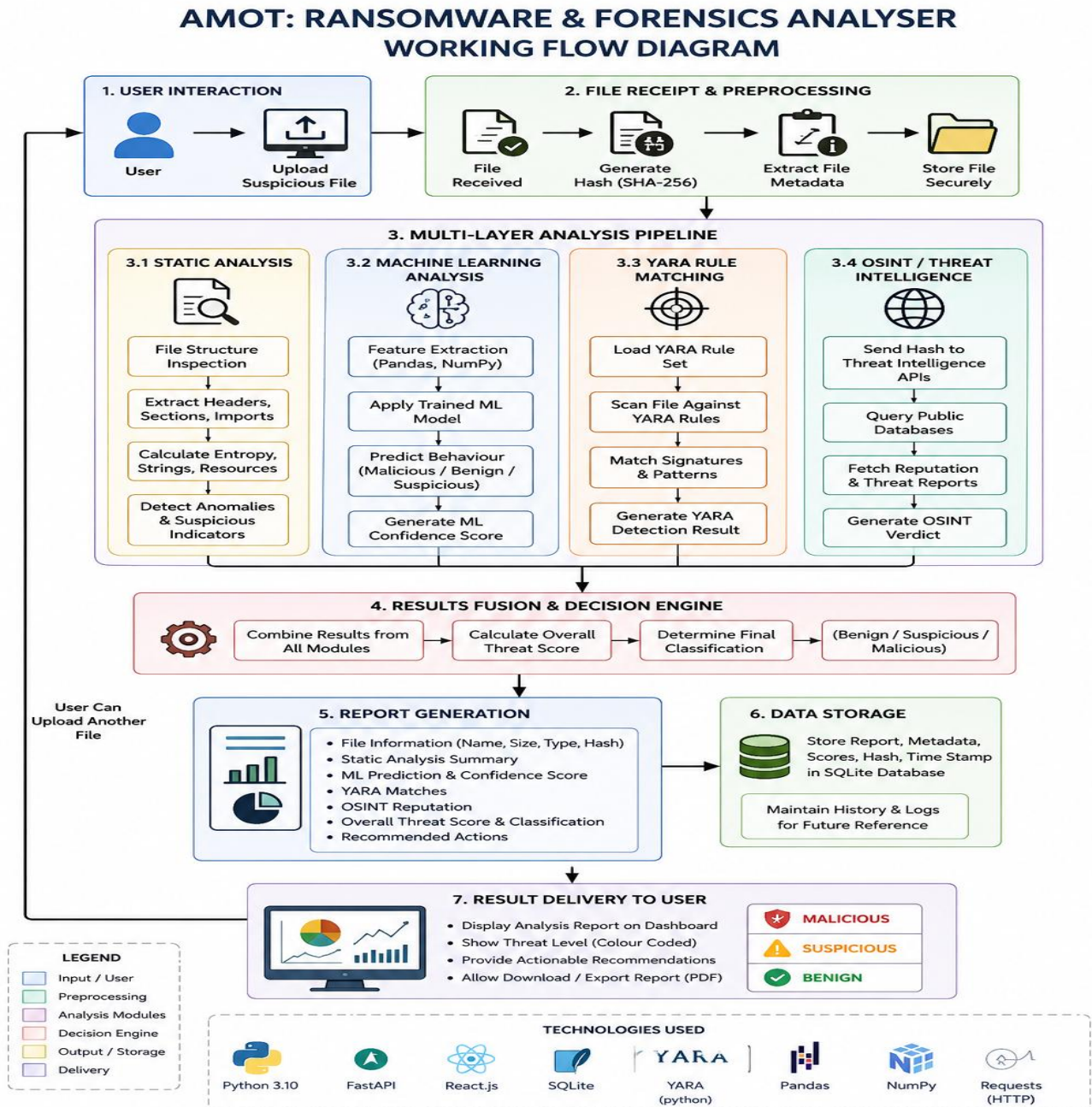
- Detailed forensic report is generated.
- Threat level and recommendations are displayed.



- Reports are stored in SQLite database.

### 3. Workflow Diagram

#### 3.1 Working Flow Diagram



#### 3.2 Workflow Explanation

1. The user uploads a suspicious file through the frontend dashboard.
2. The backend receives the file and performs preprocessing.

3. Static analysis extracts metadata, entropy values, imports, and suspicious strings.
4. Machine learning algorithms analyse extracted features to predict malicious behaviour.
5. YARA rule matching scans the file for known ransomware signatures.
6. OSINT APIs verify file reputation using cryptographic hashes.
7. Results from all modules are combined into a single threat score.
8. A detailed forensic report is generated and displayed to the user.
9. Historical analysis records are stored in the SQLite database.

## **4. System Architecture**

The architecture of the system consists of three major layers:

### **4.1 Frontend Layer**

- Developed using React.js.
- Handles file uploads and report visualization.
- Provides user dashboard and history management.

### **4.2 Backend Layer**

- Developed using Python and Fast API.
- Performs malware analysis and API handling.
- Integrates machine learning and YARA engines.

### **4.3 Database Layer**

- SQLite database stores reports and metadata.
- Maintains analysis history and logs.

### **4.4 Tools and Technologies**

| Technology  | Purpose                    |
|-------------|----------------------------|
| Python 3.10 | Backend development        |
| Fast API    | API framework              |
| React.js    | Frontend development       |
| SQLite      | Database management        |
| YARA-Python | Malware signature scanning |

|               |                       |
|---------------|-----------------------|
| Pandas        | Data processing       |
| NumPy         | Numerical analysis    |
| Requests      | OSINT API integration |
| Framer Motion | Frontend animations   |

## **5. Implementation**

The implementation phase focused on integrating all malware detection modules into a unified analytical pipeline.

The frontend dashboard allows users to upload suspicious files and view analysis reports. Once a file is uploaded, the Fast API backend processes it asynchronously through multiple analytical stages.

Static analysis modules inspect file structures and extract features. Machine learning models evaluate behavioural patterns to classify files as benign, suspicious, or malicious. YARA scanning identifies known ransomware signatures, while OSINT verification retrieves threat intelligence information from public databases.

The generated results are consolidated into a detailed forensic report displayed on the dashboard. Historical records are stored in SQLite for future investigation and auditing.

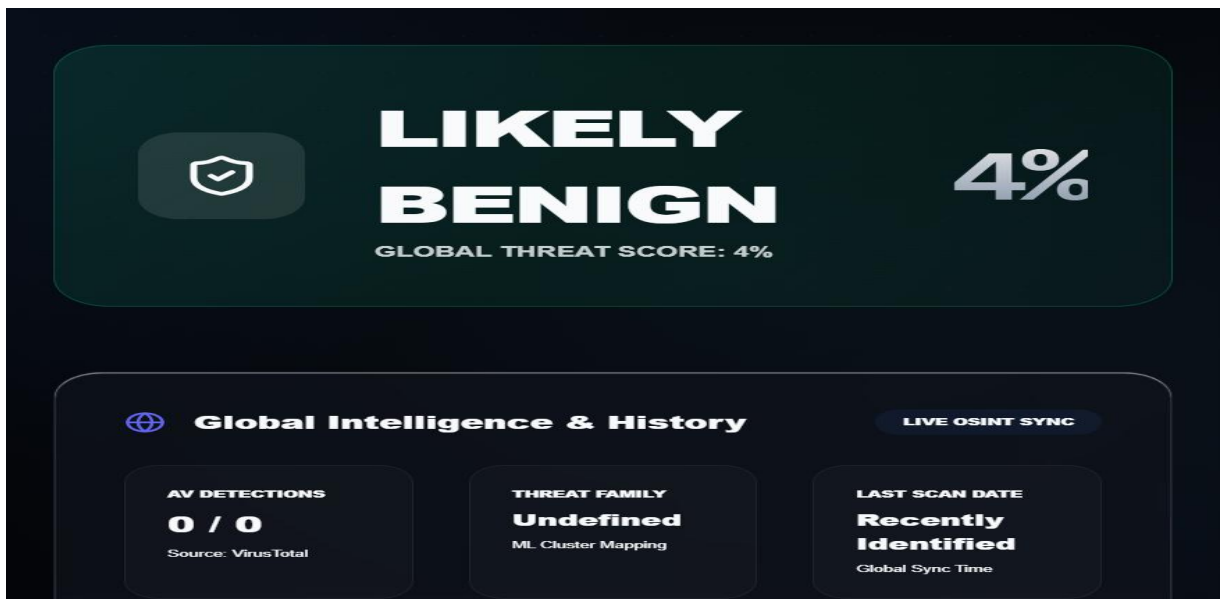
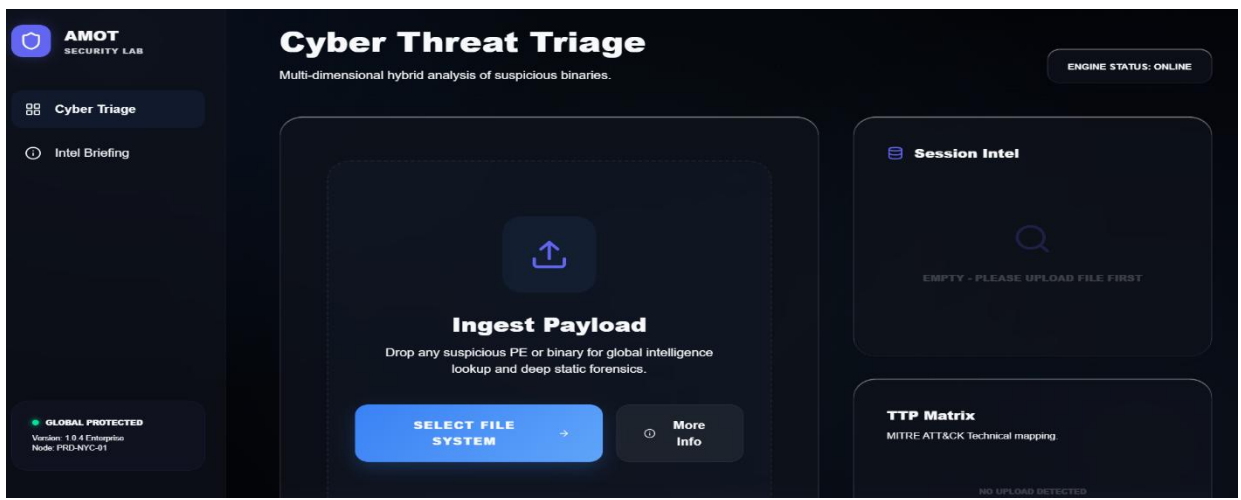
## **6. Results and Outcomes**

The AMOT platform successfully achieved its primary objectives of automating ransomware detection and improving malware analysis efficiency.

Key Outcomes

- Reduced malware investigation time.
- Improved detection accuracy.
- Reduced false positives.
- Generated detailed forensic reports.
- Detected both known and suspicious files effectively

## 6.1 Screenshots



 **Recent History**

| FILE                                                                                                     | VERDICT                                                                                                | SCORE     | DATE                   |                                                                                     |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------|------------------------|-------------------------------------------------------------------------------------|
|  ats 2.pdf              |  <b>LIKELY BENIGN</b> | <b>4%</b> | 20/3/2026, 8:50:30 am  |  |
|  <b>Final Marks.pdf</b> |  <b>LIKELY BENIGN</b> | <b>4%</b> | 19/3/2026, 10:57:12 pm |  |

 **AMOT**  
SECURITY LAB

 **Cyber Triage**

 Intel Briefing

CONSENSUS: High (Baseline Verified)  
TITANIUM-ENSEMBLE (RF + GBM + ET) | 3 MODELS SYNCED  


 **SIG MATCHES**  
0 Ransomware Rules

 **FILE IS SAFE**  
Risk assessment is low. No malicious behaviors or ransomware signatures were detected.

 **SAFETY VERIFICATION**

Our multi-stage analysis (Static, ML, and OSINT) confirms this file does not match known ransomware families. It has a valid structure and does not attempt to illegally encrypt files or hijack system processes.

## **7. CONCLUSION**

The *AMOT: Ransomware & Forensics Analyser* project successfully developed an intelligent and automated malware analysis platform capable of addressing modern ransomware threats and cybersecurity challenges. The project demonstrated how integrating multiple analytical techniques into a single framework can significantly improve malware detection accuracy, reduce false positives, and accelerate forensic investigation processes. Unlike traditional antivirus systems that rely only on signature-based detection, the proposed system combines static analysis, machine learning prediction, YARA rule matching, and OSINT-based threat intelligence verification to provide a more comprehensive and reliable approach to ransomware detection.

## **8. FUTURE SCOPE**

The system can also be expanded to include real-time network traffic monitoring and intrusion detection capabilities. This would allow the platform to analyse suspicious network activities and detect ransomware communication patterns before encryption attacks occur. Integration with SIEM (Security Information and Event Management) platforms could further strengthen enterprise-level incident response and centralized security monitoring.

Cloud deployment is another significant future scope area. Deploying the system on cloud infrastructure would improve scalability, accessibility, and performance for large-scale enterprise environments. Replacing SQLite with enterprise-grade database systems such as PostgreSQL or MongoDB would further enhance storage capacity and concurrent processing capabilities.

Additional future improvements may include automated incident response mechanisms, threat hunting modules, API-based enterprise integration, live threat intelligence feeds, and mobile dashboard support. These enhancements would transform the AMOT platform into a more advanced, enterprise-ready cybersecurity solution capable of providing proactive ransomware defence and comprehensive digital forensic analysis.

## **9. References**

1. Fast API Documentation
2. React Official Documentation
3. YARA User Guide
4. Research papers on ransomware detection and malware analysis methodologies
5. Open-source threat intelligence databases and OSINT resources
6. Python Libraries: Pandas, NumPy, Requests, SQLite3
7. Cybersecurity research articles related to machine learning in malware detection

## **10. Appendices**

The system generates and contains detailed forensic reports containing threat classification, confidence scores, matched YARA signatures, and mitigation recommendations. These reports assist cybersecurity analysts in rapid decision-making and investigation. It includes frontend dashboard screenshots, workflow diagrams, sample YARA rules, database schema details, machine learning workflow, OSINT API integration, and sample analysis reports generated during testing